SESSION – II – pl –

LIMIT THEOREMS

Weak Law of Large Numbers (WLLN)

(S. Ross, K. Trivedi)

Let $X_1, X_2, \ldots X_n$ be n iid rvs

Draw a random sample of size : $(x_1, x_2, \ldots, x_n)$

x_i is a specific value of rv X_i

* We have a common distribution of rvs having a finite mean μ and finite variance σ^2.

Sample mean $\bar{x} = \dfrac{x_1 + x_2 + \cdots + x_n}{n}$

* We would expect $\bar{x}$ will be close to μ.

Let $S_n = \sum_{i=1}^{n} X_i$, sample mean : $\bar{X} = \dfrac{\sum X_i}{n}$

$= \dfrac{S_n}{n}$

$$\operatorname{Var} \bar{X} = \operatorname{Var} \frac{S_n}{n} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

As $n \to \infty$, $\operatorname{var} \bar{X} \to 0$ (implying $\bar{X}$ is getting concentrated around mean μ). This can be further seen from application of Chebyshev's inequality.

$$P(|\bar{X} - \mu| \geq \delta) \leq \frac{\operatorname{Var} \bar{X}}{\delta^2} = \frac{\sigma^2}{n\delta^2}$$

Taking limit (W.L.L.N)

$$\lim_{n \to \infty} P(|X - \mu| \geq \delta) = 0$$

(Prob. $\bar{X}$ approximates to μ within desired accuracy δ converges to 1).

Note: The law holds under the assumption that $\{X_i\}$ have finite means.

Limit Theorems

(S. Ross – Ref).
K. Trivedi

Strong Law of Large Numbers

Let $X_1, X_2, \ldots$ be a sequence of independent rvs having a common distribution and let $E[X_i] = \mu$. Then with probability 1,

$$\frac{X_1 + X_2 + \cdots + X_n}{n} \to \mu \quad \text{as } n \to \infty$$

Ex. Suppose a sequence of independent trials is performed. Let E be a fixed event with probability P(E) that event E occurs on any particular trial.

Letting

$$X_i = \begin{cases} 1 & \text{if } E \text{ occurs on the } i\text{th trial} \\ 0 & \text{if } E \text{ does not occur} \ldots \end{cases}$$

By law of large numbers

$$\frac{X_1 + X_2 \cdots + X_n}{n} \to E(X) = P(E)$$

$X_1 + X_2 \cdots + X_n$ represents no of times E occurs

With prob. 1, the limiting proportion of time event E occurs is just P(E)

Central Limit Theorem (CLT)

Let $X_1, X_2, \dots X_n$ be a sequence of iid rvs with finite mean μ and variance σ^2.

The distribution of

$$\left[\frac{(X_1+X_2\dots+X_n)-n\mu}{\sigma\sqrt{n}}\right]$$ tends to Standard Normal

as $n \to \infty$.

This implies

$$P\left(\frac{\sum X_i - n\mu}{\sigma\sqrt{n}} \le x\right) \to \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{x} e^{-t^2/2}\,dt.$$

POWER OF THIS THEOREM : Holds for any distribution of X_i's with finite mean and variance.

Additive mechanism.

Normal Approximation to the Binomial

$X \sim Bin(n,p)$ can be considered as sum of n independent Bernoulli Variates, each with parameter p. From CLT

$$\frac{X - E(X)}{\sqrt{Var\,X}} = \frac{X-np}{\sqrt{np(1-p)}}$$ approaches standard Normal Distribution as $n \to \infty$

Note Approx. is quite good for n satisfying $np(1-p) \geq 10$.

Ex. Normal Approx. to Binomial

Let X be the number of times that a fair coin flipped 40 times lands heads. Find the probability that $X=20$. Use the normal approx. and then compare it to exact value.

Exact Result:

$$P(X=20) = \binom{40}{20}\left(\frac{1}{2}\right)^{40} \simeq \underline{\underline{0.1268}}$$

Normal Approx.

Note Binomial is a discrete r.v
Normal is a continuous rv.

It leads to a better approx. to write $P(X=20) = P(19.5 < X < 20.5)$

We have

$$P(X=20) = P(19.5 < X < 20.5)$$
$$= P\left(\frac{19.5-20}{\sqrt{10}} < \frac{X-20}{\sqrt{10}} < \frac{20.5-20}{\sqrt{10}}\right)$$
$$= P(-0.16 < Z < 0.16) \quad (1)$$
$$= \Phi(0.16) - \Phi(-0.16)$$

$$\Phi(-0.16) = P(Z > 0.16) = 1 - P(Z \le 0.16) \quad (2)$$
$$= 1 - \Phi(0.16)$$

From (1) and (2)

$$P(X=20) = \Phi(0.16) - \Phi(-0.16)$$
$$= 2\Phi(0.16) - 1 \simeq \underline{\underline{0.1272}}$$

$$\Phi(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{x} e^{-y^2/2}\,dy$$

<u>Caution</u> Don't use CLT indiscriminately

For example Cauch rv: $f(x) = \dfrac{1}{\pi(1+x^2)}$

does not finite variance.

Hence Z_n cannot be written.

Example Let x_i, $i=1,\dots 10$ be independent rvs, each being uniformly distributed over $(0,1)$. Estimate $P(\sum_{i=1}^{10} x_i > 7)$.

Soln.

$$S = x_1 + x_2 \cdots + x_{10}$$

$$E[S] = E(x_1) + \cdots + E(x_{10}) = \frac{1}{2} \times 10 = 5$$

$$V(S) = \sum_{i=1}^{10} \text{Var } x_i = 10 \times \frac{1}{12} = \frac{5}{6}.$$

$$P[S > 7] = P\left[\frac{S-5}{\sqrt{10 \cdot \frac{1}{12}}} > \frac{7-5}{\sqrt{10 \cdot \frac{1}{12}}}\right]$$

$$= P[Z > 2.2] = 1 - \Phi(2.2).$$
$$= 0.0139$$

Ex. The lifetime of a battery is a rv with mean 40 hrs and standard deviation $\sigma = 20$ hrs. Assuming a stockpile of 25 batteries, the lifetimes of which are independent, obtain the probability that over 1100 hrs of use can be achieved.

Soln

Let x_i, $i=1,\dots 25$ denote the lifetime of ith battery

$$S = x_1 + x_2 \cdots + x_{25}$$

$$E[S] = 25 \times 40 = 1000 \text{ hrs}$$
$$V[S] = \sum \text{Var } x_i = 25 \times (20)^2 = 25 \times 400 (\text{hrs})^2$$

We want

$$P(x_1 + \cdots + x_{25} > 1100) = P\left(\frac{\sum x_i - 1000}{20\sqrt{25}} > \frac{1100-1000}{20\sqrt{25}}\right)$$

$$= P(Z > 1) = 1 - \Phi(1) = 0.1587$$

Ex Let $X_1, X_2, \ldots$ be iid Poisson variates with parameter λ. Use CLT to estimate $P(120 \le S_n \le 160)$ where $S_n = X_1 + X_2 + \cdots + X_n$, $n = 75$ and $\lambda = 2$.

Soln $S_n = X_1 + X_2 + \cdots + X_n$; $X_1, X_2 \ldots X_n$ ind.

$$E[S_n] = \sum_{i=1}^{n} E[X_i] = n\lambda$$

$$V[S_n] = Var(X_1 + X_2 + \cdots + X_n) = \sum_{i=1}^{n} Var(X_i)$$
$$= n\lambda$$

Applying CLT,

$$S_n \sim N(n\lambda, n\lambda) = N(150, 150)$$

$$P(120 \le S_n \le 160) = P\left(\frac{120-150}{\sqrt{150}} < \frac{S_n - 150}{\sqrt{150}} < \frac{160-150}{\sqrt{150}}\right)$$

$$= P(-2.45 \le Z \le 0.82)$$

where

$$Z \sim N(0,1)$$

We find

$$P(120 \le S_n \le 160) = P(-2.45 \le Z \le 0) + P(0 \le Z \le 0.82)$$

$$= P(0 \le Z \le 2.45) + P(0 \le Z \le 0.82)$$

$$= 0.4929 + 0.2939 = 0.7868 \text{ (From Table)}$$

Central Limit Theorem (Proof) (Ross)

We have a sequence of iid rvs $X_1, X_2 \cdots X_n$.
Consider X_i have mean 0 and variance 1 and let common mgf be $E[e^{tX}]$

Mgf of $\dfrac{X_1 + X_2 + \cdots + X_n}{\sqrt{n}}$ $\left(\begin{matrix}\sigma = 1 \\ \mu = 0\end{matrix}\right)$.

is $E\left[e^{t\left(\frac{X_1 + X_2 + \cdots + X_n}{\sqrt{n}}\right)}\right]$

$$= E\left[e^{tX_1/\sqrt{n}}\right] E\left[e^{tX_2/\sqrt{n}}\right] \cdots E\left[e^{tX_n/\sqrt{n}}\right]$$

$$= \left\{E\left[e^{tX/\sqrt{n}}\right]\right\}^n \quad \cdots \quad (1)$$

For large n, we get

$$E\left[e^{tX/\sqrt{n}}\right] = E\left[1 + \frac{tX}{\sqrt{n}} + \frac{t^2}{2!}\frac{1}{n}X^2 + \cdots\right]$$

$$= 1 + \frac{t^2}{2n} + \cdots \quad \left(E(X) = 0,\ E(X^2) = 1\right) \quad (2)$$

Retaining terms upto $O\left(\frac{1}{n}\right)$, we have

Therefore from (1) and (2), we have

$$E\left[e^{t(X_1 + \cdots X_n)/\sqrt{n}}\right] \approx \left(1 + \frac{t^2}{2n}\right)^n \quad (3)$$

As $n \to \infty$, we have

$$\lim_{n \to \infty} \left(1 + \frac{t^2}{2n}\right)^n = e^{t^2/2}$$

Therefore

$$\lim_{n\to\infty} E\left[e^{t(X_1+X_2+\cdots+X_n)/\sqrt{n}} \right] = e^{t^2/2}$$

We find the mgf of $\frac{X_1+X_2+\cdots+X_n}{\sqrt{n}}$ converges to mgf of a standard normal variate $N(0,1)$.

When X_i have mean μ and variance σ^2, the rv $Y_i = \frac{X_i-\mu}{\sigma}$ have mean 0 and variance 1.

Proceeding on similar lines, we can show that

$$P\left\{ \frac{(X_1-\mu)+(X_2-\mu)\cdots+(X_n-\mu)}{\sigma\sqrt{n}} \le a \right\} \to \Phi(a)$$

This proves C.L.T.

<u>CLT</u> : Thm. Let $X_1, X_2, \ldots, X_n$ be independent rvs with a finite mean $E[X_i] = \mu_i$ and finite variance $Var\, X_i = \sigma_i^2$ $(i = 1, 2, \ldots n)$.
We form the normalized rv

$$Z_n = \frac{\sum_i X_i - \sum_i \mu_i}{\sqrt{\sum_i \sigma_i^2}}$$

so that $E[Z_n] = 0$ and $Var\, Z_n = 1$.

Under certain regularity conditions, the limiting distribution of Z_n is standard normal,
$Z_n \to N(0,1)$.

$$\lim_{n \to \infty} F_{Z_n}(t) = \lim_{n \to \infty} P(Z_n \le t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{t} e^{-y^2/2}\, dy.$$

REVIEW

CLT tells us sums of independent rvs tends to normally distributed.

Theorem: Let $X_1, X_2, \ldots, X_n$ be independent rvs with finite mean $E[X_i] = \mu_i$ and a finite variance $V(X_i) = \sigma_i^2$ $(i = 1, 2, \ldots n)$.

Form normalized rv

$$Z_n = \frac{\sum_i X_i - \sum_i \mu_i}{\sqrt{\sum \sigma_i^2}},$$

so that

$$E[Z_n] = 0, \quad V(Z_n) = 1.$$

Under certain regularity conditions, the limiting distribution of Z_n is standard normal,

i.e. $Z_n \to N(0,1)$

$$\lim_{n \to \infty} F_{Z_n}(t) = \lim_{n \to \infty} P(Z_n \le t) = \int_{-\infty}^{t} \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy.$$

Ex. <u>Special Case of CLT.</u>

Assume $X_1, X_2, \ldots X_n$ are mutually independent and identically distributed with common mean $\mu = E[X_i]$ and common variance $Var(X_i) = \sigma^2$. Then

$$Z_n = \frac{(\bar{X} - \mu)}{\sigma / \sqrt{n}} = \frac{(\bar{X} - \mu)\sqrt{n}}{\sigma} \to N(0,1)$$